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Editorial

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1 Preface

Knowledge graphs (KGs; Hogan et al., 2021) have become essential for realizing the Semantic Web vision, enabling the development of intelligent, data-driven systems and lately supporting advancements in large language models (LLMs). Yet, its construction remains a complex and evolving challenge that continues to attract attention from both academia and industry (Chaves-Fraga et al., 2024). This special issue brings together recent research contributions that advance the state of the art in this field, offering theoretical insights, methodological innovations, and practical solutions that collectively push the boundaries of KG construction (KGC) and pave the way for future innovations.

Over the past decade, significant progress has been made in developing methods and systems for transforming semistructured data, such as tabular data (e.g., relational databases, or CSV files), or hierarchical data (e.g., XML, or JSON) into resource description framework (RDF)-based KGs (Van Assche et al., 2023). Declarative frameworks, including mapping languages (Dimou et al., 2014; Iglesias-Molina et al., 2023) and rule-based approaches, have laid the foundations for transparent and reproducible KGC (Arenas-Guerrero et al., 2024; Iglesias et al., 2020). However, important challenges remain regarding the formalization of mapping languages, the automation of mapping rules definition, the scalability and performance of KGC systems, as well as their systematic evaluation. Addressing these issues requires both conceptual advances and efficient implementations that can cope with the increasing volume and heterogeneity of data.

The human factor introduces its own complexities as well, including designing intuitive and accessible user interfaces, minimizing cognitive load, and balancing automation with manual oversight. Equally demanding is the process of conducting user-centered evaluations, which poses challenges in defining meaningful metrics, capturing diverse user needs, and ensuring that systems are both usable and effective in real-world scenarios. Addressing these issues requires incorporating human-in-the-loop approaches and enabling iterative feedback to ensure that the resulting systems meet real-world usability.

In parallel, the field is undergoing a transformation driven by machine learning (ML) and, more recently, LLMs (Freund et al., 2025; Schmidt et al., 2025). These approaches are opening new avenues for the automation and enrichment of KGs (Dimou & Chaves-Fraga, 2022), from schema generation and entity extraction to data validation and quality assessment. While such models bring remarkable potential, they also raise new questions concerning explainability, reproducibility, and human oversight, calling for hybrid approaches that integrate declarative methods with ML techniques in a human-in-the-loop setting.

The contributions included in this special issue reflect these diverse but interconnected perspectives. They explore the interplay between declarative and procedural paradigms, the optimization of mapping systems, and the integration of ML

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and LLMs in the knowledge engineering lifecycle. Together, they offer a comprehensive and up-to-date picture of the research landscape, illustrating how established methodologies continue to evolve while new paradigms emerge.

This collection provides both an overview of the current state of KGC and inspiration for further research in this vibrant and rapidly advancing area of the Semantic Web.

2 Contributions

This special issue received 14 articles, among which seven articles were accepted for publication. Among the accepted articles, one article is related to mapping rules' formalization (Oo et al., 2025), two articles to optimizations of KGC systems (García-González, 2025; Van Assche et al., 2025), three articles to LLMs (Blin et al., 2025; Dang et al., 2025; Moskvoretskii et al., 2025), two articles to the event-centric KGC (Blin et al., 2025; Van Assche et al., 2025). and one article to users' explainability of the KGC process (Zhang et al., 2025). In detail:

- *Viktor Moskvoretskii, Irina Nikishina, Ekaterina Neminova, Alina Lobanova, Alexander Panchenko, and Chris Biemann. Large language models for creation, enrichment and evaluation of taxonomic graphs* (Moskvoretskii et al., 2025). This paper investigates the use of LLMs for automating taxonomy-related tasks, such as construction, enrichment, and evaluation of taxonomic graphs. The authors introduce TaxoLLaMA, a unified model fine-tuned on datasets derived from WordNet 3.0, capable of addressing a broad range of lexical semantic tasks. Experimental results show that the model achieves state-of-the-art performance across multiple benchmarks, demonstrating the potential of LLMs for learning and refining taxonomic relations with high accuracy.
- *Bohui Zhang, Albert Meroño-Peñuela, and Elena Simperl. Towards explainable automated knowledge engineering with human-in-the-loop* (Zhang et al., 2025). This paper addresses the growing opacity of modern KGC pipelines, which increasingly depend on black-box ML models and heterogeneous data sources. The authors propose a framework for explainable, human-in-the-loop knowledge engineering, combining a systematic literature review with interviews from domain experts. Their study identifies key use cases and gaps in explainability, outlines user requirements, and proposes design blueprints to improve transparency, accountability, and fairness in automated KGC processes, aligning them with emerging artificial intelligence governance and regulatory frameworks.
- *Minh-Hoang Dang, Thi Hoang Thi Pham, Pascal Molli, Hala Skaf-Molli, and Alban Gaignard. LLM4Schema.org: Generating Schema.org markups with large language models* (Dang et al., 2025). This paper presents LLM4Schema.org, a novel framework for evaluating the ability of LLMs to generate valid and semantically accurate Schema.org markup for web content. Unlike traditional benchmarking methods, LLM4Schema.org compares LLM-generated annotations directly with human-produced markup, without requiring predefined ground truth. The study reveals that a significant portion of LLM outputs remain invalid or non-compliant, highlighting current limitations in ontology adherence. However, the authors show that LLM-based validation agents can effectively filter and improve the quality of generated markup, with GPT-4 ultimately surpassing human performance after correction, demonstrating both the promise and the challenges of using LLMs for large-scale semantic annotation.
- *Inès Blin, Ilaria Tiddi, Remi van Trijp, and Annette ten Teije. ChronoGrapher: Event-centric knowledge graph construction via informed graph traversal* (Blin et al., 2025). This paper introduces ChronoGrapher, a system designed to automatically construct event-centric KGs from large, generic graphs such as DBpedia or Wikidata. The approach combines a semantically guided, best-first traversal to extract event-relevant subgraphs with a hybrid enrichment strategy that integrates structured and text-based information. ChronoGrapher demonstrates strong adaptability across datasets and outperforms existing methods in building coherent, temporally grounded event graphs. A user study further shows that incorporating event-centric triples significantly improves the factual grounding and relevance of responses in event-related question answering tasks.
- *Herminio García-González. Optimising the ShExML engine through code profiling: From turtle's pace to state-of-the-art performance* (García-González, 2025). This paper focuses on improving the performance of the ShExML engine, a user-friendly language for KGC. Through detailed code profiling and optimization, the author identifies and resolves key performance bottlenecks, followed by a rigorous statistical evaluation of the enhancements. The optimized engine achieves execution times comparable to other state-of-the-art mapping systems, significantly improving its scalability and responsiveness. As a result, ShExML now provides a more efficient and reliable solution for users constructing KGs from heterogeneous data sources.
- *Sitt Min Oo, Ben De Meester, Ruben Taelman, and Pieter Colpaert. Algebraic mapping operators for knowledge graph generation* (Oo et al., 2025). This paper introduces a formal algebraic framework for defining the operational semantics of KG generation processes. Building upon the SPARQL algebra, the proposed algebraic mapping operators provide a unified foundation for representing and optimizing mappings across different languages, such

as RML and ShExML. The authors demonstrate that this approach enables language-independent optimization and fosters consistency between mapping and query engines. Experimental results show stable, low-memory performance, with the prototype ranking highly in the KGC Workshop's performance challenge. This work establishes a solid theoretical basis for analyzing the complexity, expressiveness, and optimization of KG mapping systems.

- *Dylan Van Assche, Julian Rojas, Ben De Meester, and Pieter Colpaert. Incremental knowledge graph construction from heterogeneous data sources* (Van Assche et al., 2025). This paper presents IncRML, an incremental approach to KG generation that efficiently handles evolving and dynamic data sources. Built on top of RML and FnO, the method detects and materializes data changes—creates, updates, and deletes—while minimizing redundant processing. Detected changes can be published as Linked Data Event Streams using the W3C Activity Streams 2.0 vocabulary, enabling semantic communication of updates over the Web. Experimental results across multiple real-world datasets demonstrate substantial reductions in storage, central processing unit time, and memory usage, achieving up to 315× lower storage costs and 4.4× faster construction. The approach significantly lowers the cost of maintaining up-to-date KGs and promotes scalable, Web-native publication of evolving data.

3 Analysis and Beyond

After analyzing the accepted articles, their main contributions focus on two areas: the development of standards and formalization techniques for declarative methods for KGC, and the automation or provision of user support throughout the KGC process.

3.1 Standards and Standardization

A recurring theme in KGC is the role of standards and their adoption by the community. In this special issue, we received contributions related to the RML (Dimou et al., 2014; Iglesias-Molina et al., 2023) and ShExML (García-González et al., 2020) mapping languages. RML extends R2RML,[1] a W3C Recommendation for transforming data contained in relational databases into RDF. RML has been instrumental in supporting mappings beyond relational databases. The KGC Community Group[2] was established in 2019 and has been actively working toward establishing a charter for a candidate submission to the W3C since 2023. Similarly, ShExML builds upon ShEx,[3] which, although not a W3C Recommendation, is widely acknowledged as a specification for validation by the Semantic Web community. This close alignment with recognized and widely adopted specifications underscores how ShExML, such as RML, is rooted within standards, whether identified by a standardization body or community.

We note that this special issue did not receive contributions on Façade-X (Asprino et al., 2024; Daga et al., 2021). Yet, it is important to recognize that a W3C Community Group[4] was established in September 2025 for this initiative as well. Façade-X enables the use of SPARQL to interact with various sources as RDF graphs. RML and ShExML differ from Façade-X in that the former focus on declaratively specifying mappings from source data to RDF according to a target ontology. In contrast, the latter focuses on using SPARQL to access data contained in heterogeneous data sources as RDF via direct mappings. These direct mappings are a natural evolution of a direction mapping approach as proposed by the *Direct Mapping of Relational Data to RDF*[5] and generalized to different data sources by Façade-X. With Façade-X, one can thus use SPARQL CONSTRUCT to generate RDF graphs using different vocabularies and create KGC pipelines.

Taken together, these developments suggest that emerging standards are evolving to address different use cases and niches in KGC. However, the different approaches highlight both the richness of the field and the importance of continued dialogue between communities to ensure maximum interoperability and sustainable adoption.

3.2 Mapping Languages' Formalization

Another observation is the increased effort within the community to formalize the mapping languages for the KGC. While R2RML introduced a reference algorithm, it did not provide a formal foundation. This gap has been addressed by, for example, Calvanese et al. (2017) and Elhalawati et al. (2025). The first offered a formalization of R2RML for the Ontop system; the second proposed a Datalog-based formalization of R2RML. In this special issue, Min Oo et al. presented initial steps toward formalizing RML. Although this work was well received, we note for the readers' benefit that the lead author of the article has recently published a similar work in collaboration with a computer scientist (Oo & Hartig, 2025), which won the Best Paper award at the 22nd European Semantic Web Conference (ESWC) in 2025.

We expect that the formalization of mapping languages will play an important role in shaping and driving the standardization process of KGC. Historically, KGC has been led primarily by engineers focused on system development and practical applications. In contrast, formalization is often the domain of (theoretical) computer scientists. As KGC

languages grow more expressive and begin to intersect with fields such as programming languages, logic, and database theory, closer collaboration between these communities will be essential. It will otherwise be challenging to establish solid, extensible, and verifiable foundations.

3.3 Large Language Models (LLMs)

A notable trend is the integration of LLMs to streamline and enhance various stages of KGC, ranging from initial design to iterative refinement and enrichment. This relationship between LLMs and KGs is bidirectional. On the one hand, LLMs can support and accelerate the KGC; on the other hand, the continuous enhancement of KGs with newly constructed knowledge strengthens the performance of LLMs, improving their factual grounding, reducing hallucinations, and enabling more accurate reasoning. LLMs have been considered so far for all aspects of KGC; ranging from ontology development and taxonomy construction and enrichment, as with (Moskvoretskii et al., 2025), to table interpretation and KG enrichment. However, as automated approaches increasingly rely on LLMs, challenges arise regarding the validity and interpretability of the results, as Dang et al. (2025) showed. Ambiguous or incorrect results highlight the need for mechanisms for their explainability, as with (Zhang et al., 2025), validation, as with (Dang et al., 2025), and improvement are required.

Moreover, as KGC is a complex process that requires a good understanding of the data to semantically annotate entities and their relations, we observe that KGC with declarative mapping languages remains primarily manual. Automated systems for the KGC rely more and more on LLMs, but either only indicate the semantic annotations without constructing a knowledge graph, or directly produce a knowledge graph without producing any declarative mapping rules. As with declarative KGC communities that have their own community groups and venues, for example, the KGC workshop,[6] the automated KGC topic has its own communities, for example, around the SemTab challenge[7] or the Table Representation Learning Workshop.[8] Consequently, we observe that the declarative and automated approaches evolve in different directions, and little interaction occurs between them.

3.4 Human-in-the-Loop

Although human interactions for KGC was one of the highlighted topics of the special issue, only one article addressed this aspect. Existing approaches for human interactions tend to fall into two categories: declarative methods, which emphasize editor tools and structured methodologies such as the pay-as-you-go paradigm (Sequeda & Miranker, 2017) to assist users, and automated solutions that often rely on human feedback for refinement. However, there has been limited investment in comprehensive human-in-the-loop methodologies that actively integrate user input throughout the entire construction process. Recognizing this gap, a dedicated workshop on "Users and Knowledge Graphs"[9] was recently organized to foster dialogue, share best practices, and promote research that prioritizes usability, collaboration, and human-centered design in knowledge graph development.

4 Challenges and Future Work

KGC continues to evolve rapidly, driven by new paradigms, technologies, and the increasing need for scalable, explainable, and interoperable data integration methods. Despite the significant advances reflected in this special issue, several open challenges remain and point toward promising directions for future research.

A first challenge lies in bridging declarative and learning-based approaches. Declarative mapping languages provide transparency, reproducibility, and formal rigor, whereas ML and LLMs offer adaptability and automation. The integration of both paradigms, through neuro-symbolic approaches with human-in-the-loop pipelines, remains an open area that requires methodological frameworks and evaluation metrics capable of balancing interpretability and performance.

A second line of research concerns evaluation and benchmarking. Although several benchmarks such as KROWN (Van Assche et al., 2024) and GTFS-Madrid-Bench (Chaves-Fraga et al., 2020) have played an important role in assessing the performance of KGC engines, they require updates to align with current languages, specifications, and parameters. Moreover, the rise of LLM-based and incremental construction approaches calls for new evaluation scenarios that go beyond traditional materialization pipelines. Recent initiatives such as the BLINKG benchmark (Castedo et al., 2026) represent a significant step toward reproducible, fine-grained, and comparable evaluation of automatic KGC methods. However, broader community coordination is still needed to establish unified datasets, metrics, and evaluation protocols that ensure fairness and cumulative scientific progress.

Another emerging challenge involves maintaining and evolving KGs in dynamic and distributed environments. The growth of data spaces and the proliferation of real-time data streams call for incremental and event-driven graph construction methods, enabling continuous synchronization, provenance tracking, and version management while minimizing

computational costs (Geisler et al., 2025). Finally, from a foundational perspective, formalization and standardization remain essential. Ongoing efforts toward algebraic semantics for mapping languages and formal models of execution need to be consolidated (Oo & Hartig, 2025) and connected with W3C standardization initiatives. Ensuring semantic interoperability across languages, engines, and specifications will be key to achieving maturity and long-term sustainability in the field.

In summary, the future of KGC lies in combining solid theoretical foundations with adaptive, explainable, and interoperable systems. The convergence of declarative design, formal semantics, and ML holds the potential to make KGC not only more efficient and scalable but also more trustworthy and impactful across scientific and industrial domains.

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Notes

1. https://www.w3.org/TR/r2rml/
2. https://www.w3.org/community/kg-construct/
3. https://shex.io/
4. https://www.w3.org/groups/cg/fx/
5. https://www.w3.org/TR/rdb-direct-mapping/
6. http://w3id.org/kg-construct/workshop
7. https://www.cs.ox.ac.uk/isg/challenges/sem-tab/
8. https://sites.google.com/view/eurips25-ai-td/home
9. https://ukgworkshop.github.io

References

Arenas-Guerrero, J., Chaves-Fraga, D., Toledo, J., Pérez, M. S., & Corcho, O. (2024). Morph-KGC: Scalable knowledge graph materialization with mapping partitions. *Semantic Web*, *15*(1), 1–20.

Asprino, L., Daga, E., Dowdy, J., Gangemi, A., & Mulholland, P. (2024). Materialisation approaches for façade-based data access with SPARQL. *Semantic Web: – Interoperability, Usability, Applicability*. Accepted

Blin, I., Tiddi, I., van Trijp, R., & ten Teije, A. (2025). Chronographer: Event-centric knowledge graph construction via informed graph traversal. *Semantic Web*, *16*(5), 22104968251377247. https://doi.org/10.1177/22104968251377247

Calvanese, D., Cogrel, B., Komla-Ebri, S., Kontchakov, R., Lanti, D., Rezk, M., Rodriguez-Muro, M., & Xiao, G. (2017). Ontop: Answering SPARQL queries over relational databases. *Semantic Web*, *8*(3), 471–487. https://doi.org/10.3233/SW-160217

Castedo, C., Iglesias, E., Lama, M., Bugarín-Diz, A., Vidal, M. E., & Chaves-Fraga, D. (2026). BLINKG: A benchmark for LLM-integrated knowledge graph generation. *Transactions on Graph Data and Knowledge*. Accepted.

Chaves-Fraga, D., Corcho, O., Dimou, A., Vidal, M. E., Iglesias-Molina, A., & Van Assche, D. (2024). Are knowledge graphs ready for the real world? Challenges and perspective. *Dagstuhl Reports*, *14*(2), 1–70.

Chaves-Fraga, D., Priyatna, F., Cimmino, A., Toledo, J., Ruckhaus, E., & Corcho, O. (2020). GTFS-Madrid-Bench: A benchmark for virtual knowledge graph access in the transport domain. *Journal of Web Semantics*, *65*, 100596.

Daga, E., Asprino, L., Mulholland, P., & Gangemi, A. (2021). Facade-X: An opinionated approach to SPARQL anything. In M. Alam, P. Groth, V. de Boer, T. Pellegrini, H. J. Pandit, E. Montiel-Ponsoda, V. Rodríguez-Doncel, B. McGillivray & A. Meroño-Peñuela (Eds.), *Further with knowledge graphs—proceedings of the 17th international conference on semantic systems, SEMANTiCS 2021, Amsterdam, The Netherlands, September 6–9, 2021, studies on the semantic web* (Vol. 53, pp. 58–73). IOS Press. https://doi.org/10.3233/SSW210035

Dang, M. H., Pham, T. H. T., Molli, P., Skaf-Molli, H., & Gaignard, A. (2025). LLM4Schema.org: Generating Schema.org markups with large language models. *Semantic Web: – Interoperability, Usability, Applicability*, *16*(6). https://doi.org/10.1177/22104968251382172

Dimou, A., & Chaves-Fraga, D. (2022). Declarative description of knowledge graphs construction automation: Status & challenges. In *Proceedings of the 3rd international workshop on knowledge graph construction (KGCW 2022) co-located with 19th extended semantic web conference (ESWC 2022)*.

Dimou, A., Vander Sande, M., Colpaert, P., Verborgh, R., Mannens, E., & Van de Walle, R. (2014). RML: A generic language for integrated RDF mappings of heterogeneous data. In C. Bizer, T. Heath, S. Auer, & T. Berners-Lee (Eds.), *Proceedings of the workshop on linked data on the web co-located with the 23rd international world wide web conference (WWW 2014), Seoul, Korea, April 8, 2014*, *CEUR workshop proceedings* (Vol. 1184). CEUR-WS.org. https://ceur-ws.org/Vol-1184/ldow2014_paper_01.pdf.

Elhalawati, A., Van den Bussche, J., & Dimou, A. (2025). A declarative formalization of R2RML using datalog and its efficient execution. In *Companion proceedings of the 9th international joint conference on rules and reasoning (RuleML + RR 2025) also co-located with 21th reasoning web summer school (RW 2025) and 17th DecisionCAMP 2025 as part of declarative AI 2025, Istanbul, Türkiye, September 22–24, 2025, CEUR workshop proceedings* (Vol. 4083). CEUR-WS.org. https://ceur-ws.org/Vol-4083/paper63.pdf.

Freund, M., Dorsch, R., Schmid, S., & Harth, A. (2025). Mapping by example: Towards an RML mapping reverse engineering pipeline. In *Proceedings of the 6th international workshop on knowledge graph construction (KGCW 2025) co-located with 22nd extended semantic web conference (ESWC 2025)*.

García-González, H., Boneva, I., Staworko, S., Gayo, J. E. L., & Lovelle, J. M. C. (2020). ShExML: Improving the usability of heterogeneous data mapping languages for first-time users. *PeerJ Computer Science*, *6*, e318. https://doi.org/10.7717/peerj-cs.318

García-González, H. (2025). Optimising the ShExML engine through code profiling: From turtle's pace to state-of-the-art performance. *Semantic Web 16*, (2), SW–243736. https://doi.org/10.3233/SW-243736

Geisler, S., Cappiello, C., Celino, I., Chaves-Fraga, D., Dimou, A., Iglesias-Molina, A., Lenzerini, M., Rula, A., Van Assche, D., Welten, S., & Vidal, M. E. (2025). From genesis to maturity: Managing knowledge graph ecosystems through life cycles. *Proceedings of the VLDB Endowment*, *18*(5), 1390–1397.

Hogan, A., Blomqvist, E., Cochez, M., d'Amato, C., Melo, G. D., Gutierrez, C., Kirrane, S., Gayo, J. E. L., Navigli, R., Neumaier, S., Ngomo, A. C., Polleres, A., Rashid, S. M., Rula, A., Schmelzeisen, L., Sequeda, J., Staab, S., & Zimmermann, A. (2021). Knowledge graphs. *ACM Computing Surveys (Csur)*, *54*(4), 1–37.

Iglesias, E., Jozashoori, S., Chaves-Fraga, D., Collarana, D., & Vidal, M. E. (2020). SDM-RDFizer: An RML interpreter for the efficient creation of RDF knowledge graphs. In *Proceedings of the 29th ACM international conference on information & knowledge management* (pp. 3039–3046). ACM.

Iglesias-Molina, A., Van Assche, D., Arenas-Guerrero, J., De Meester, B., Debruyne, C., Jozashoori, S., Maria, P., Michel, F., Chaves-Fraga, D., & Dimou, A. (2023). The RML ontology: A community-driven modular redesign after a decade of experience in mapping heterogeneous data to RDF. In T. R. Payne, V. Presutti, G. Qi, M. Poveda-Villalón, G. Stoilos, L. Hollink, Z. Kaoudi, G. Cheng, & J. Li (Eds.), *The semantic web—ISWC 2023—22nd international semantic web conference, Athens, Greece, November 6–10, 2023, proceedings, part II, lecture notes in computer science* (Vol. 14266, pp. 152–175). Springer. https://doi.org/10.1007/978-3-031-47243-5_9

Moskvoretskii, V., Neminova, E., Lobanova, A., Panchenko, A., & Nikishina, I. (2025). Large language models for creation, enrichment and evaluation of taxonomic graphs. *Semantic Web: – Interoperability, Usability, Applicability*, *17*(1). https://doi.org/10.1177/22104968251404186

Oo, S. M., De Meester, B., Taelman, R., & Colpaert, P. (2025). Algebraic mapping operators for knowledge graph generation. *Semantic Web*, *16*(5), 22104968251361350. https://doi.org/10.1177/22104968251361350

Oo, S. M., & Hartig, O. (2025). An algebraic foundation for knowledge graph construction. In E. Curry, M. Acosta, M. Poveda-Villalón, M. van Erp, A. K. Ojo, K. Hose, C. Shimizu & P. Lisena (Eds.), *The semantic web—22nd European semantic web conference, ESWC 2025, Portoroz, Slovenia, June 1–5, 2025, proceedings, part I, lecture notes in computer science* (Vol. 15718, pp. 3–22). Springer. https://doi.org/10.1007/978-3-031-94575-5_1

Schmidt, W. J., Grangel-González, I., Huschle, T., Wagner, L., Kharlamov, E., & Paschke, A. (2025). LLM-supported mapping generation for semantic manufacturing treasure hunting. In *European semantic web conference* (pp. 84–101). Springer.

Sequeda, J. F., & Miranker, D. P. (2017). A pay-as-you-go methodology for ontology-based data access. *IEEE Internet Computing*, *21*(2), 92–96.

Van Assche, D., Chaves-Fraga, D., & Dimou, A. (2024). KROWN: A benchmark for RDF graph materialisation. In *International semantic web conference* (pp. 20–39). Springer.

Van Assche, D., Delva, T., Haesendonck, G., Heyvaert, P., De Meester, B., & Dimou, A. (2023). Declarative RDF graph generation from heterogeneous (semi-) structured data: A systematic literature review. *Journal of Web Semantics*, *75*, 100753.

Van Assche, D., Rojas, J., De Meester, B., & Colpaert, P. (2025). Incremental knowledge graph construction from heterogeneous data sources. *Semantic Web: – Interoperability, Usability, Applicability*, *17*(2). https://doi.org/10.1177/22104968251412270

Zhang, B., Meroño-Peñuela, A., & Simperl, E. (2025). Towards explainable automated knowledge engineering with human-in-the-loop. *Semantic Web*, *16*(5), 22104968251382171. https://doi.org/10.1177/22104968251382171